

A2 Level Physics (Edexcel IAL) — Formula Sheet

Edexcel International A2: Units 4–5 (WPH14/15). Everything from the AS sheet is assumed knowledge.

UNIT 4 — FURTHER MECHANICS, FIELDS & PARTICLES

Momentum 2D	conserve p per axis; $p = mv$
Impulse	$\text{impulse} = F \cdot t = \Delta(mv)$ (N·s) — area under F-t graph
Circular motion	$\dot{e} = v/r = 2\pi/T$; $a = v^2/r = \dot{e}^2 r$; $F = mv^2/r$
Angular vs linear	$v = \dot{e} r$
Coulomb's law	$F = q \cdot q / (4\pi\epsilon_0 r^2)$ (point charges)
Electric field	$E = F/q$; radial: $E = q / (4\pi\epsilon_0 r^2)$; uniform: $E = V/d$
Electric potential	$V = q / (4\pi\epsilon_0 r)$; $E = -\nabla V$
Capacitance	$C = Q/V$ (F, C, V)
Capacitor energy	$E = \frac{1}{2}QV = \frac{1}{2}CV^2 = Q^2/2C$
Capacitor discharge	$Q = Q_0 e^{-(t/RC)}$; time constant $\tau = RC$
Magnetic force	$F = BIL \cdot \sin\theta$ (wire); $F = Bqv \cdot \sin\theta$ (charge)
Magnetic flux	$\Phi = BA \cdot \cos\theta$ (Wb); flux linkage $N\Phi$
Faraday & Lenz	$\mathcal{E} = -N \cdot d\Phi/dt$
Transformer	$V_s/V_p = N_s/N_p$ (ideal: 100% efficient)
Particle physics	$E = mc^2$; $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$; $E = hc/\lambda$ for photon

UNIT 5 — THERMODYNAMICS, RADIATION, OSCILLATIONS

Ideal gas	$pV = NkT = nRT$ ($k = 1.38 \times 10^{-23} \text{ J/K}$)
Kinetic theory	$\frac{1}{2}m\langle c^2 \rangle = (3/2)kT$ (mean kinetic energy of a molecule)
Internal energy	= sum of random kinetic + potential energies of molecules
Specific heat	$\Delta E = mc\Delta\theta$
Specific latent heat	$\Delta E = mL$ (fusion: solid $\rightarrow$ liquid; vaporisation: liquid $\rightarrow$ gas)
Thermal conductivity	$Q/t = kA \cdot \Delta\theta/x$ (k : W/(m·K))
Blackbody radiation	$P = e\sigma A T^4$ $\sigma = 5.67 \times 10^{-8} \text{ W/(m}^2\text{K}^4)$
Radioactive decay	$A = \lambda N$; $N = N_0 e^{-(\lambda t)}$; $A = A_0 e^{-(\lambda t)}$
Half-life	$t_{1/2} = \ln 2 / \lambda \approx 0.693 / \lambda$
SHM	$x = A \cdot \cos(\dot{e}t)$; $a = -\dot{e}^2 x$; $\dot{e} = 2\pi f$
SHM velocity	$v = \pm \dot{e} \sqrt{A^2 - x^2}$; max at centre: $v = A\dot{e}$
Mass-spring	$T = 2\pi \sqrt{m/k}$
Pendulum	$T = 2\pi \sqrt{L/g}$ (small angles)

ASTROPHYSICS & NUCLEAR (option topics)

Gravitational field	$g = F/m$; $g = GM/r^2$ (radial)
Gravitational potential	$V = -GM/r$ (zero at infinity)
Wien's law	$\lambda_{\text{max}} \cdot T = 2.898 \times 10^{-3} \text{ m} \cdot \text{K}$
Standard candles	intensity $\propto 1/d^2$ (inverse square) — $I = L/(4\pi d^2)$
Hubble	$v = H_0 d$ (expansion of the universe)
Nuclear radius	$R = r_0 \cdot A^{1/3}$ $r_0 \approx 1.05 \text{ fm}$

KEY CONSTANTS (given)

μ_0 , k , σ , h	$8.85 \times 10^{-12} \text{ F/m}$; $1.38 \times 10^{-23} \text{ J/K}$; $5.67 \times 10^{-8} \text{ W/(m}^2\text{K}^4)$; $6.63 \times 10^{-34} \text{ J} \cdot \text{s}$
e , m_e , m_p	$1.60 \times 10^{-19} \text{ C}$; $9.11 \times 10^{-31} \text{ kg}$; $1.67 \times 10^{-27} \text{ kg}$